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# CLUES: Community-driven LoRa networks Urban sensing Empowered by Smart homes

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***Abstract*—Long-range and low-power networks play a crucial role in the development of smart cities. However, the centralized and vertically integrated Internet of Things (IoT) model, where municipalities or private entities deploy and manage devices effectively retaining ownership of the data, presents significant technological and operational barriers that have so far hindered the capillary deployment of the technology. To overcome these limitations, we propose an original bottom-up deployment model that decentralizes data and network ownership and reduces management and operational barriers. Community-driven LoRa networks Urban sensing Empowered by Smart homes (CLUES) advocates for a participatory and decentralized sensor network through the integration of a smart gateway into citizen's smart homes. Crowd-sensed data captured by Smart Home Gateways (SHG) allows us to power local-scoped digital representations of sensors, augmenting their capabilities with data and services from edge devices. Gateways can participate in municipal networks and share sensed data with institutions and companies, fostering a cost-effective approach to the construction of city ecosystems. In particular, our proposal opens up new deployment and management models for low-power networks, providing better coverage and enabling more dense, scalable, and energy-efficient deployments. We showcase the effectiveness of our approach through a large-scale evaluation of a simulated deployment of the proposal and provide the community with an open-source implementation of the SHG.**

***Index Terms*—Low Power Wide Area Networks; Participatory Sensing; Data Decentralization; LoRa Network Densification.**

## I. Introduction

Modern cities are increasingly required to evolve and integrate smarter behaviors to improve sustainability, resilience, and accessibility for citizens. Sensors and IoT devices deployed throughout the city play a crucial role in this mission, continuously collecting data to enable informed decision-making, smarter resource management, and enhanced citizen interactions [1]. Particularly important for this objective are the low-power and cost-effective sensors that utilize long-range communication technologies such as Long Range (LoRa), allowing the deployment of affordable devices even in challenging environments, such as areas with limited power supply or network coverage [2].

Generally, this sensing infrastructure follows a centralized deployment model, maintained by municipalities or private entities, which create services that eventually expose the data to citizens and other stakeholders. However, this *modus operandi* presents substantial challenges for new initiatives [1]. In fact, the combination of high initial investment, uncertain revenue models, and the complexity of navigating a fragmented ecosystem makes monetization difficult, making it harder for new stakeholders to justify investment and maintain long-term growth in the sector.

In this work, we present an original bottom-up deployment model that integrates commodity LoRa devices into citizen's smart homes. In our reference scenario, smart homes actively participate in data gathering and management processes, transparently incorporating sensors into their systems via standardized technology. This can be achieved through the Smart Home Gateway (SHG), which acts as a bridge between physical sensors, the smart home, and the city. The SHG embodies a pluggable architecture that allows to seamlessly interface locally scoped data with internal and external services, enabling the creation of scalable and value-added services for the community. In contrast to other participatory LoRaWAN models like [3], [4], where gateways act solely as packet forwarders and rely on centralized backend infrastructure, the SHG architecture enables a decentralized paradigm, allowing to distribute data and network management responsibilities across a network of SHGs and edge services, thereby reducing reliance on a single point of control.

Community-driven LoRa networks Urban sensing Empowered by Smart homes (CLUES) embraces denser and more scalable network deployments with better or comparable performance to that of standalone Long Range Wide Area Network (LoRaWAN) technology, while retaining the benefits of lower network deployment and management costs. At the same time, the SHG can easily integrate with or act as a bridge to other standards and technologies, making it a suitable candidate also for rural or less densely populated environments. We argue that this new approach can foster new forms of cooperation, allowing citizens to control and obtain timely and locally-scoped information from shared sensors and systems. In line with the principles of open science, we provide an open-source implementation of the SHG, accessible in [5], to facilitate its adoption by the wider research and industrial community.

The remainder of the article is organized as follows: Section II provides a concise overview of the LoRa technology and improvements, differentiating our work from existing models. Section III introduces CLUES and a prototype implementation of the SHG, while Sec. IV is dedicated to the experimental evaluation. Finally, in Sec. V we draw our conclusions and outline directions for future work.

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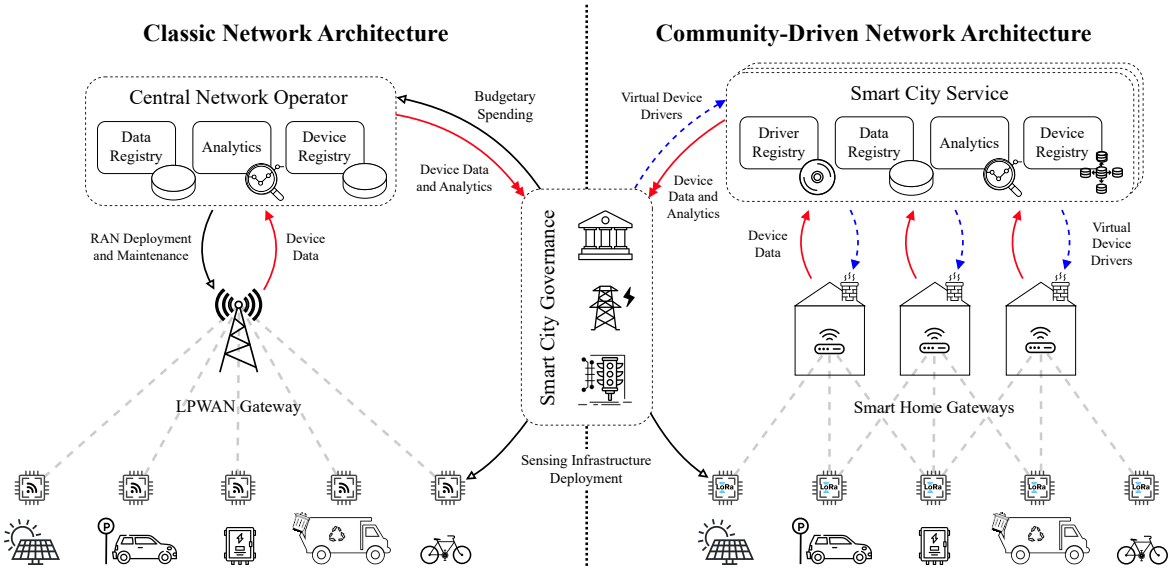



Fig. 1. On the left side is shown a single cell in a classic LPWAN architecture [3], while on the right is shown our participatory, decentralized LoRa network. Compared to a classic architecture, our Smart Home Gateways can be installed into citizens homes to foster a densification of receivers while providing participants direct access to locally sensed information and relieving municipalities from the cost of deploying and managing the infrastructure.

## II. Low Power Long Range Sensing for the Smart City

Low Power Wide Area Networks (LPWANs) are a class of IoT technologies designed to support sensing infrastructures in large-scale environments, relying on battery-powered sensing and energy-harvesting devices, not depending on the electrical grid. To enhance the longevity of these networks, minimizing energy consumption during network transmission is paramount [6].

LPWAN technologies can be classified into two categories based on the spectrum. The first group operates on licensed spectra (e.g., NB-IoT, LTE-M, and EC-GSM-IoT), while the second operates on unlicensed Industrial, Scientific and Medical (ISM) bands (e.g., LoRa, Sigfox, Wize, and MIOTY) [3].

Compared to cellular LPWANs, the adoption of unlicensed solutions in Smart Cities offers numerous advantages: (i) they operate in free sub-gigahertz or 2.4GHz radio frequency bands, contributing to their affordability and flexibility; (ii) they do not require SIM cards and heavy authentication procedures; and (iii) they adopt a much simpler Medium Access Control (MAC) layer, aimed at fully maximizing the battery lifetime of intermittently and sporadically active devices [6].

Among the several technologies proposed by research and industry, LoRa has emerged in the market due to its cost-effectiveness, easy-to-operate nature, and better power efficiency. LoRa is a proprietary wireless physical layer technology based on Chirp Spread Spectrum (CSS) modulation, whose key feature is the ability to support concurrent transmissions on a shared channel by varying the Spreading Factor (SF) of the symbols, leveraging the quasi-orthogonality among spreading factors to reduce interference. In addition, higher SFs reduce data rates, by improving receiver sensitivity and by extending the transmission range.

### A. LoRa-Based LPWAN Sensor Networks

LoRa only defines the lower physical layer in the communication stack, allowing for the use of different upper layer network protocols, with LoRaWAN being by far the most popular. LoRaWAN adopts a star-of-stars topology [3]: as shown on the left-hand side of Fig. 1, end devices transmit messages to one or more base stations also called *Gateways*, which encapsulate and forward sensed data to a *Network Server* via a cellular or other IP network. The Central Network Operator harvests sensed data from gateways and exposes them to co-located and remote smart city services, implementing storage and analytics services for citizens, municipalities, and businesses.

The LoRaWAN MAC layer defines three operative classes for devices (A, B, and C), where class A is the most energy efficient and adopted one. Class A devices always initiate uplink transmission of data on a random channel without prior coordination; conversely, downlink communication from the network operator can occur only in one of two reception slots scheduled by the device after transmission. This simple operation allows devices to spend most of their time in a sleep state for energy preservation purposes, resulting in minimal energy consumption.

LoRaWAN is known to deliver limited performance in dense environments due to its ALOHA-like nature [2]. Indeed, sensor communications can be subject to network congestion and interference from collision with other devices' transmissions, registering heavy packet loss rates. To overcome issues arising from uncoordinated access to the medium, many techniques have been proposed in the literature, advocating access control and retransmission mechanisms at the higher levels of the protocol stack [7]. Examples of these techniques include time-slotting and channel-sensing mechanisms that prevent transmission collisions. However, these enhancements introduce additional overhead, contributing to an increase in the energy consumption of the sensors, possibly undermining the key

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goal of LPWAN technology [2]. These problems are further exacerbated in dense deployment scenarios, where collision statistically increases and synchronization protocols rise in complexity as well.

### B. Toward Participatory LoRa Networks

A promising emerging approach is to exploit a denser distribution of LoRa receivers to improve the overall reliability of the network and data reception. The densification of receiver devices increases the likelihood of data transmission reception, enabling devices to use a lower SFs. This reduces per-packet airtime, decreasing the overall probability of packet collisions [2]. In this context, device-based relays have been proposed to enable multi-hop transmissions over LoRa [8]. These relays can be used to build mesh topologies and extend coverage in both urban and remote areas, potentially reducing energy consumption and improving reliability [4], [9], [10]. However, LoRa-to-LoRa relays and mesh topologies also come at the cost of limited scalability in the presence of heavy traffic [8]. With CLUES, we propose a complementary approach to improve network capacity which takes advantage of cheap sensing devices and gateway (access network) densification.

Moreover, most existing proposals rely on a central network server, thus not achieving a fully distributed architecture. For example, the authors in [11] propose to distribute server functionality using a gateway federation, providing scalable roaming and improved contention management. A key drawback of the centralized model is that a single entity handles all network maintenance and is in charge of its long-term sustainability, including the deployment and maintenance of base stations, the hosting of the central server and the management of utility services. This role could be fulfilled directly by municipalities and other smart city service providers or offloaded to network operators. In both cases, the cost of managing the infrastructure falls on the service providers, possibly hindering the capillary and large-scale deployment of the technology.

Recent initiatives, such as The Things Network and Helium [4], aim to move toward a densification and decentralization of LoRa deployments. In this context, participants voluntarily deploy and maintain gateways, motivated either by direct personal or business interests, or by incentives such as financial remuneration. Recognizing these previous efforts, CLUES advances the field by promoting a wider participation of smart city citizens in the formation of sensor networks in the city. In contrast to The Things Network and Helium, where gateways act as mere data forwarders and network management operations are centralized, CLUES embodies a pluggable architecture that supports gateway virtualization and custom drivers, allowing the decentralization of data and management processes. CLUES emphasizes functional integration, with incentives arising naturally from access to value-added services.

Depicted on the right-hand side of Fig.1, CLUES advocates a new sensor network architecture where gateways are integrated into citizen smart homes, acting both as contextual information sources for homeowners and as information providers for the community as a whole. CLUES embraces network densification, both gateways and sensing devices, leading to decentralization of services and data, enabling easier access to smart city information.

## III. A Participated and Distributed Approach for LoRa Sensor Networks

CLUES aims at the creation of a participatory LoRa-based network, decentralizing the data collection effort, allowing the sharing and distribution of data through an integrated ecosystem of smart home gateways and smart city services. In the following, we sketch the core elements of CLUES, successively providing some practical insight into the implementation of a prototype, whose source code can be consulted in repository [5].

### A. Smart Home Gateway Architecture & Services

The core element of CLUES is the *Smart Home Gateway*, which enables data collection from LoRa and LoRaWAN-compliant sensors. Support for LoRaWAN is mandatory, considering it being a de facto industrial standard protocol. The gateways, pictorially depicted in Fig. 2, are installed by citizens in their homes, incentivized by the possibility of a direct and free acquisition of valuable data from their surroundings, such as the status of bins, pollution and temperature/humidity sensors, etc. Data collected by SHG can be processed and stored locally, and can be exposed as interactive information to the local community and/or shared with the broader smart city through northbound infrastructure and services. This capability also enables access to raw and aggregated information for smart city governance, which can take advantage of it for planning actions in the city.

SHGs continuously listen for incoming LoRa/WAN transmissions. Sensed data are stored in a reception buffer part of the Hardware Abstraction Layer (HAL), a pluggable software component that offers send/receive/configuration primitives to upper-layer functionalities (see Sec.III-B). Buffered data are inspected and successively routed for further processing, stored, and are eventually exposed to the smart home and/or to the broader CLUES network.

*Virtual Devices* (VDs) are trusted programs running within the SHG environment, acting mainly as adapters to integrate LoRa devices and data into the smart home and smart city ecosystem of services. The virtual device has a modular architecture, designed as a set of loosely coupled modules. The *Device Driver* is the only mandatory component in our CLUES implementation as it integrates essential business logic and knowledge, e.g., data schema, certificates, and encryption/decryption keys, which are mandatory in case of LoRaWAN-compliant sensors to receive and deserialize data from the associated devices. Additional modules like Analytics, Data Fusion, and Exposure enable VDs to cooperate within the broader participatory network and offer homeowners advanced functionalities over the collected data.

The *Data Fusion Module* handles the synchronization of current and historical data with the SHG Storage Service and smart city Data Registries. The Storage Service is an endpoint

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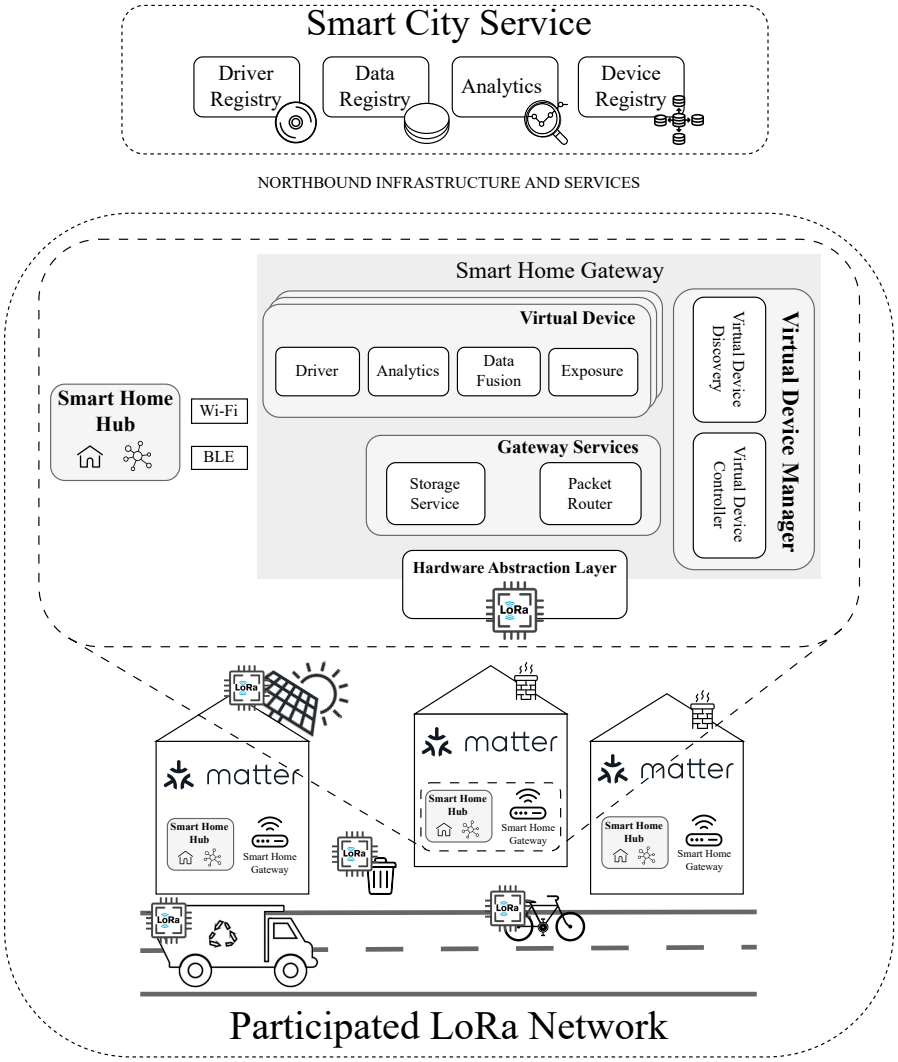



Fig. 2. Pluggable software architecture of the Smart Home Gateway capable of dynamic loading of device specific drivers (Virtual Devices) handling data synchronization, (pre)processing and exposure to local and/or global services. The SHG can interact with smart city services via northbound interfaces, and integrate locally with Smart Home products by exploiting standard Matter interfaces.

available in each SHG that allows virtual devices to store and retrieve information. Information can be fetched by analytics functions or provided as a response to an external request routed via the Exposure Module. On the other front, smart city Data Registries are synchronization and memorization functions, deployed by smart city communities or sensor providers, acting as global network endpoints for contextual data collection. Historic information collected in the Data Registries can be exploited, and sources can be correlated to infer behaviors at different time and geographical granularities. As an additional security feature, cross-validation during data aggregation enables services to detect inconsistencies and malicious SHG nodes.

The *Analytics Module* consists of utility functions that VDs use to process data transmitted by a sensor. A single VD can integrate multiple such modules to answer different business objectives. As an example, a VD associated with a traffic sensor can take advantage of new and past values to provide street traffic trends and predictions to the owner of the Smart Home (SH). To answer more computationally intensive tasks, the Analytics Modules can offload part of the computational burden to remote Analytic Services provided by the smart city.

Finally, the *Exposure Module* is capable of responding to internal and external queries and distribution of information in the SH network. External interactions leverage north-bound interfaces routed via the global network, while internal ones can integrate with local services, e.g., existing SH Hubs such as Samsung Smart Things, Amazon Alexa, or Google Nest, etc.

The management plane is tasked with essential management functionalities, including VD lifecycle management through the *Virtual Device Controller* (VDC) and the discovery and dynamic loading of virtual devices from external repositories through the *Virtual Device Discovery*. The current design choice is to have a one-to-one mapping between sensors and virtual devices, allowing for the creation of augmented digital representation of sensors.

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### B. A Smart Home Gateway Prototype

We have developed a prototype of the SHG exploiting affordable commercial off-the-shelf hardware to promote broader adoption. The SHG can be deployed on any X86/ARM low-cost board, such as a Raspberry Pi, equipped with LoRa transceiver of the SX127x family. These chips can receive a single channel and a single spreading factor at a time, and the choice is driven by the significant reduction in hardware costs for participants, which facilitates greater adoption by homeowners [12]. To interface with this transceiver, the HAL exposes a configuration endpoint, which allows the dynamic runtime configuration of the channel and the spreading factor, along with the necessary send/receive primitives for communication with the upper layers.

The dynamic transceiver reconfiguration is implemented to maximize the replication factor of transmitted data, which affects network resiliency, minimizing packet loss across the network. The goal is to ensure that each transmitted packet is received by at least one SHG. This capability helps overcome the best-effort nature of the classic LoRa protocol, increasing the density of listening points in the network.

The current design relies on one of the two channel access strategies: Random and Single-Channel SF7 (SCSF7). In the Random strategy, devices can arbitrarily choose which carrier frequency and SF is used to transmit data without any agreement with the network. SHGs randomly switch the frequency and spreading factor they listen to. In this context, a high penetration rate of SHGs provides statistical guarantees that each transmission channel within the area is covered, while posing no design constraint on the sensor side.

The SCSF7 strategy assigns a fixed carrier frequency and SF7 to all devices and SHGs to transmit and receive the packet. This method alleviates the need for network synchronization and offers great flexibility and support in high-density and mobile scenarios. However, it prevents sensors from using different spreading factors and frequencies to improve distribution or overcome environmental obstacles. By minimizing the transmission range, this strategy maximizes data locality and reduces network interference, allowing denser sensor deployments.

On the software side, once the SHG is deployed, the HAL periodically polls the reception buffer, and whenever a new packet is detected, it is forwarded through a Linux pipe to a lightweight demultiplexing functionality. This component strips the device ID following a LoRaWAN MAC header (MHDR), and the data is published to the matching message-oriented middleware. The design choice to adopt the LoRaWAN MHDR for both LoRa and LoRaWAN devices is to provide uniform support for both sensor types, delegating more complex functionalities to individual device drivers running within the VDs. The HAL differentiates among the two sensor types by relying on the information embedded in the Frame Type (FType = 111) field. This capability is required as LoRaWAN data packets need to be routed to the correct VD, tasked also with payload data decryption. Several design options emerge to fully support LoRaWAN-compliant network functions, ranging from the deployment of a roaming-capable network server in the SHG, to its distribution and integration between smart home and smart city services [6].

TABLE I
SIMULATION PARAMETERS.

| | |
|---|---|
| Physical topology | Uniform device placement, randomized over a 84.64km$^2$ area (maximum distance of 13km as in [13]) |
| Radio channel model | Log-distance path loss with Gaussian shadow fading [13]; capture-effect and SF quasi-orthogonality on collision interference [2] |
| Sources of packet loss (by precedence) | Received signal strength under sensitivity; mismatch in SF and carrier frequency; busy receiver; interference from collision |
| Traffic pattern | Periodic, 255B/20 minutes (close to worst case scenario with SF12 under the 1% duty cycle fair use policy of the EU 868MHz band [2]) |
| Network runtime | 10 hours |
| Frequency channels | 125 kHz bandwidth around 8 central carrier frequencies in the EU 868MHz band [2] |
| MAC strategies | Random (8 channels, SF=7-12, TP=14dBm), SCSF7 (1 channel, SF=7, TP=14dBm), LoRaWAN (8 channels, ADR) |
| Randomized runs | 30 with different seed per unique setting, 95% confidence intervals in plots |

ZeroMQ is used as a low-overhead and efficient middleware to perform asynchronous, non-blocking forwarding of data to VDs for improved scalability in constrained environments. If the stripped device ID does not have a matching route, the Virtual Device Discovery fetches information about the device via an external source, relying on the northbound infrastructure and services interfaces. To this end, public or private stakeholders can choose to run their own Device Registry or participate in a federated, unified registry that spans multiple vendors.

Another task performed by the VDC is to configure a ZeroMQ topic for the newly onboarded sensor, where the VD subscribes to receive sensed data. To enable an efficient and low-footprint storage service, VDs rely on a shared instance of RocksDB, instantiated by the VDC which regulates access to the resource. VDs implement a default basic behavior in which, as Matter-compliant smart devices, they expose the collected data within the smart home environment. For the source code and additional details on the implementation of the SHG, we refer the reader to [5].

## IV. PARTICIPATED LORA NETWORK PERFORMANCE

Without loss of generality, we evaluate the feasibility of our proposal by focusing on a large-scale deployment of the solution. To this end, we implemented the protocol stack and a simplified SHGs in the well-known simulator *ns-3* [14], while a field evaluation of the SHG is planned for the near future.

The objective is to assess (i) whether CLUES provides any assurance of data reception, that is, the data transmitted from the sensors are received from at least one SHG and (ii) the eventual degree of sensor data delivery, evaluated by counting, for each received packet, the average number of gateways able to obtain a copy of the data packet. In the latter experiment, we indirectly assess the resiliency of the system in the face of potential local disruptions. Finally, we estimate the energy consumption for each configuration, as it is a critical factor that

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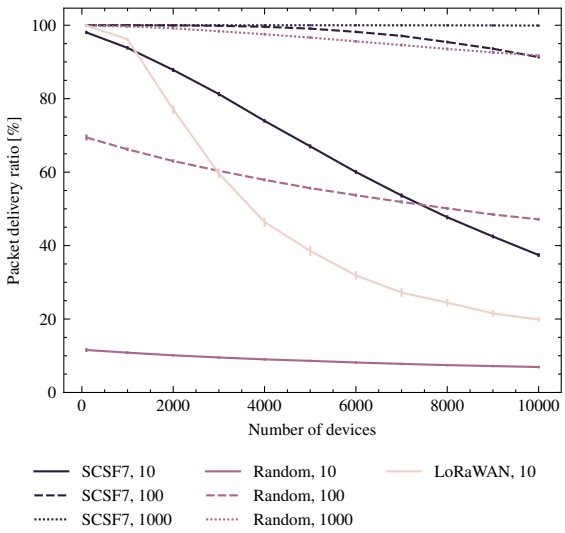



Fig. 3. Network packet delivery ratio for an increasing number of devices. Labels indicate the scenario and the number of deployed gateways. LoRaWAN's number of gateways is limited to 10 to match a city-wide deployment size [13].

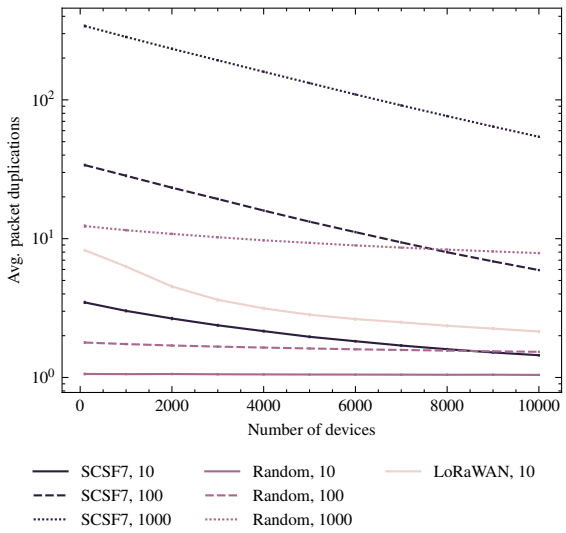



Fig. 4. Average number of duplicated receptions per successful packet for an increasing number of devices. Labels indicate the scenario and the number of deployed gateways. LoRaWAN's number of gateways is limited to 10 to match a city-wide deployment size [13].

influences the maintenance and adoption costs of low-power devices.

The physical layer of the sensor and the gateway are modeled on the SX1272 transceiver [15]. We evaluate two different configurations of our SHG network, namely the SCSF7 and the Random strategy. In the SCSF7 configuration, both SHGs and devices exclusively use SF7 on a common 125KHz channel. In the Random configuration, they independently pick a combination of carrier frequency and SF.

We compare the aforementioned LoRa strategies against a LoRaWAN deployment, consisting of gateways equipped with a multichannel SX1301 transceiver. The gateways have 8 parallel reception paths, the devices randomly hop over 8 frequencies for each new transmission and have had their SF and Transmission Power (TP) lowered by an Adaptive Data-Rate (ADR) strategy [3]. In our case, ADR is statically reproduced by lowering SF and TP to ensure a transmission outage probability lower than 2.275% ($2\sigma$ above the mean of path loss) from the closest gateway. Each transmission is evaluated for correct reception at each gateway instantiated in the simulation. Additional details and simulation can be found in Table I.

### A. Network Packet Delivery Ratio

We measure the Packet Delivery Ratio (PDR) metric, obtained as the ratio between the total number of packets received by at least one gateway, over the total number of packets sent in the simulation. The PDR is a widely used metric in the literature whose importance is twofold: in low-density scenarios, it can be used to evaluate access network coverage (i.e., how much of the traffic is in reception range); in good coverage, high channel contention settings, it shows the network's ability to scale with the number of collisions caused by more devices.

PDR values are reported in Fig. 3 for an increasing number of devices. In general, PDR always degrades to interference when the number of devices increases drastically. This can be mitigated by exploiting the LoRa capture effect by increasing the density of SHG. The results indicate that 10 SCSF7 SHGs can already cover the deployment area for a low-density deployment (more than 95% PDR), while 1000 (12 gw/km$^2$) can maintain over 99% PDR up to 10.000 devices (118 dev/km$^2$).

To put things in perspective, the city of Bologna in Italy has 1470 houses/km$^2$ over an extension of 141km$^2$, therefore the number of smart-home gateways to support an even bigger network should be easily achievable in an urban area. Moreover, we test a worst-case scenario for the traffic profile of devices, thus the number we provide should be considered a lower bound.

### B. Local Data Availability

In this experiment, we evaluate the ability of our proposal to decentralize the data collection process. As an added advantage, a greater duplication factor of sensor data increases the redundancy of the deployment to make it more reliable. To this end, we trace the average number of recipients for each successful packet in the simulation, as shown in Fig. 4.

Overall, the SCSF7 approach scores better than the Random one, as the latter transmissions may be additionally lost due to a lack of receivers tuned to the correct SF and channel. Compared to a city-wide LoRaWAN deployment of 10 multi-channel gateways [13], duplications are lower for 10 SHGs. This holds until we factor in the cost of infrastructure, as classic gateway hardware is more expensive [12]. For instance, multi-channel gateway chips in the EU are generally an order of magnitude more expensive than single-channel ones. Thus, considering that the performance of SCSF7 over 100 SHGs is discretely higher, we gather that our solution might directly reach more locations compared to a LoRaWAN deployment.

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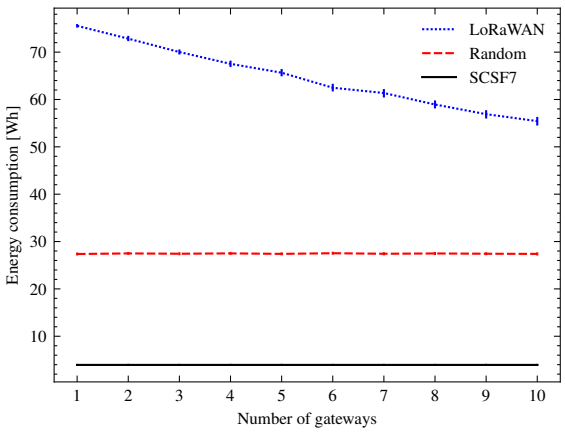



Fig. 5. Total energy consumption [Wh] of 10000 (i.e. 118 dev/km$^2$) LoRa transceivers for an increasing number of gateways. SCSF7 and Random are compared to a plausible LoRaWAN urban deployment, usually not larger than of 10 gateways [13].

Increasing the number of devices in the deployment conveys the effects of interference: duplications decline exponentially, likely due to the combination of the log-distance path loss model with the threshold-based LoRa capture effect model in the simulation. Thus, on average, receivers of a transmission will be the ones physically closer to the origin device. This also aligns with the design principle that the closer the source, the more valuable the data is for the receiver.

### C. Energy Consumption of Devices

Energy consumption is a key factor in determining the lifetime and maintenance costs of low-power devices. We estimate the energy consumption [Wh] of the LoRa transmitters using the following energy model. LoRa equipment can operate in 4 main power modes (Transmit, Standby, Receive, and Idle/Sleep) with a respective current draw of 28mA, 1.4mA, 10.8mA, and 1.5$\mu$A/0.1$\mu$A [15]. The simulation precisely traces the amount of time spent by devices in each power state which, assuming a constant 3.3V supply tension, enables energy estimation.

In our case, devices only shift between Idle and Transmit states. Note that higher SFs yield exponentially longer transmission durations, which affects energy consumption. The values in [15] also indicate that, during transmission, the LoRa chip draws 28mA of current to output 13dBm of radio power and 18mA to produce 7dBm. A linear interpolation of these values allows us to model current values for the different TP settings used by LoRaWAN.

The energy consumption values are shown in Fig. 5. SHG and Random produce constant values at 0.2Wh and 1.4Wh, respectively. In the LoRaWAN scenario, energy consumption starts at 3.8Wh with one gateway and decreases to 2.7Wh with 10 gateways. This behavior can be traced to the parameter allocation done by LoRaWAN's ADR algorithm, which would eventually converge to the power consumption of SCSF7 for nonviable amounts of multi-channel gateways. We conclude that a solution such as SCSF7 can significantly reduce the energy consumption of devices compared to LoRaWAN.

## V. Conclusion

A key enabler for smarter cities is the widespread deployment of low-cost, low-power communication technologies capable of supporting large-scale sensing and actuation. In this context, we presented a novel approach that integrates LoRa networks into smart homes by leveraging the capabilities of the SHG. Our proposal helps simplify the deployment and management of LoRa networks, while promoting a participatory model in which citizens actively contribute to and benefit from smart city data. CLUES has the potential to support a wide range of local and global-scoped smart city services, such as efficient energy management in energy communities, timely flood risk alerts, optimized waste collection, etc., while also enabling flexible, scalable, and energy-efficient deployments. In the future, we plan to extend our solution to ensure full compliance with the LoRaWAN specification, enabling seamless integration with existing IoT infrastructures, fostering broader adoption. CLUES paves the way for heterogeneous deployments, which encompass diverse LoRa devices and management approaches, highlighting the need for future work on incentive mechanisms to foster cooperation and encourage positive stakeholder behavior.

## References

[1] B. M. Chan *et al.*, "IoT Advisory Board Report," National Institute of Standards and Technology (NIST), U.S. Dept. of Commerce, Oct. 2024, Accessed: Feb. 10, 2025. [Online]. Available: https://www.nist.gov/system/files/documents/2024/10/21/The%20IoT%20of%20Things%20Oct%202024%20508%20FINAL_1.pdf

[2] P. Gkotsiopoulos, D. Zorbas, and C. Douligeris, "Performance determinants in LoRa networks: A literature review," *IEEE Commun. Surv. & Tut.*, vol. 23, no. 3, pp. 1721–1758, Jun. 2021.

[3] C. Milarokostas, D. Tsolkas, N. Passas, and L. Merakos, "A comprehensive study on LPWANs with a focus on the potential of LoRa/LoRaWAN systems," *IEEE Commun. Surv. & Tut.*, vol. 25, no. 1, pp. 825–867, Dec. 2022.

[4] D. Wu and J. Liebeherr, "A low-cost low-power LoRa mesh network for large-scale environmental sensing," *IEEE Internet of Things J.*, vol. 10, no. 19, pp. 16 700–16 714, Apr. 2023.

[5] CLUES. Mobile Middleware Research Group. Accessed: Feb. 10, 2025. [Online]. Available: https://github.com/MMw-Unibo/CLUES

[6] M. Jouhari, N. Saeed, M.-S. Alouini, and E. M. Amhoud, "A survey on scalable LoRaWAN for massive IoT: Recent advances, potentials, and challenges," *IEEE Commun. Surv. & Tut.*, vol. 25, no. 3, pp. 1841–1876, May 2023.

[7] Z. Sun, H. Yang, K. Liu, Z. Yin, Z. Li, and W. Xu, "Recent advances in LoRa: A comprehensive survey," *ACM Trans. Sensor Netw.*, vol. 18, no. 4, pp. 1–44, Nov. 2022.

[8] J. R. Cotrim and J. H. Kleinschmidt, "LoRaWAN mesh networks: A review and classification of multihop communication," *Sensors*, vol. 20, no. 15, p. 4273, Jul. 2020.

[9] M. S. Aslam *et al.*, "Exploring multi-hop LoRa for green smart cities," *IEEE Netw.*, vol. 34, no. 2, pp. 225–231, Oct. 2019.

[10] B. Arratia, E. Rosas, C. T. Calafate, J.-C. Cano, J. M. Cecilia, and P. Manzoni, "AILoRa: Empowering environmental intelligence through an advanced LoRa-based IoT solution," *Comput. Commun.*, vol. 218, pp. 44–58, Mar. 2024.

[11] S. Delbruel, N. Small, E. Aras, J. Oostvogels, and D. Hughes, "Tackling contention through cooperation: A distributed federation in LoRaWAN space," in *Proc. of EWSN*, 2020, p. 13–24.

[12] C. Zhong and X. Nie, "A novel single-channel edge computing LoRa gateway for real-time confirmed messaging," *Scientific Reports*, vol. 14, no. 1, p. 8369, Apr. 2024.

[13] M. Rademacher, H. Linka, T. Horstmann, and M. Henze, "Path loss in urban LoRa networks: A large-scale measurement study," in *Proc. IEEE Veh. Technol. Conf.*, 2021, pp. 1–6.

[14] LoRaWAN ns-3 module. SIGNET Research Group. Accessed: Oct. 25, 2024. [Online]. Available: https://apps.nsnam.org/app/lorawan

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[15] Semtech Corp., Wireless & Sensing Products Division, Camarillo, CA, USA. *SX1272/73 Datasheet*, Rev. 4 (2019). Accessed: Oct. 25, 2024. [Online]. Available: https://www.semtech.com/products/wireless-rf/lora-connect/sx1272

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